MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Paper Code : BCAC601 Unix and Shell programming

UPID : 600105

Time Allotted : 3 Hours *Full Marks :70*

The Figures in the margin indicate full marks.

Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer any ten of the following : [1 x 10 = 10]

(I) What are the categories of processes in UNIX? Name them.

(II) Which command is used to sort the lines of data in a file in reverse order

(III) List two essential tasks of system administration.

(IV) How do we know which shell is running for us?

(V) What is UNIX command to display the pathname of the current directory?

(VI) What happens when more than one filenames are given with cat command?

(VII) How to see the calendar for month July 2023?

(VIII) Name the file that stores the pathname of the home directory in UNIX.

(IX) Which is the command to copy an entire directory structure in UNIX?

(X) What is the default permissions for regular files in UNIX?

(XI) How to record your login session?

(XII) Sometimes, the system refuses to create a directory using mkdir command. Why?

Group-B (Short Answer Type Question)

Answer any three of the following : [5 x 3 = 15]

2. Briefly explain the UNIX file system. [5]
3. Briefly explain the following commands: (i) cp, (ii) cat [5]
4. What do you mean by ordinary, directory and device files in UNIX? What are text file and binary files in UNIX? [5]
5. How you can delete files in UNIX? [5]
6. Explain the use of tail in UNIX with suitable examples. [5]

Group-C (Long Answer Type Question)

Answer any three of the following : [15 x 3 = 45]

7. (a) Give a diagram of UNIX architecture. [2]
   (b) What is kernel and shell? [6]
   (c) How to print a calendar for a particular month and year? [2]
   (d) How to find out who has logged in? [2]
   (e) What is purpose of man and cat command? [2]
   (f) Give an example of internal command in UNIX. [1]
8. (a) Explain with suitable diagram the parent – child relationship of UNIX file system. [7]
   (b) Briefly describe /etc and /dev files of UNIX. [4]
   (c) Write the significance of dot (.) and dotdot (..) in UNIX. [4]
9. (a) How to display the current date and time in UNIX? [2]
   (b) Give the UNIX command to display only the present month. [2]
   (c) Provide the command to display the date in mm/dd/yy format. [2]
   (d) Give the UNIX command to display (i) current hour, (ii) current minute, and (iii) current second. [6]
   (e) How to display (i) machine's characteristics, (ii) name of the platform, and (iii) machine's hardware name? [3]
10. (a) List the commands which are used to do the followings: (i) display the pathname of the current directory, (ii) changing the current directory, (iii) create directory, (iv) remove directory, (v) listing [10]

directory, (vi) list all files including hidden files, (vii) list file's inode index number, (viii) to change directory to the home directory, (ix) to move to the parent directory of current directory, and (x) get the details of directory information.

(b) Briefly explain /bin, and /lib. [5]

11. (a) Why you use chown, chgrp command? [6]

(b) Write the commands using relative permission for the followings: (i) Deny execute permission to everyone, (ii) Allow read permission to everyone, and (iii) Make a file readable and writable by the group and others. [6]

(c) How you can set permissions to the files recursively? [3]

*** END OF PAPER ***

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